Octagonal Turn Counter — At-the-Bench Build Guide

Scope: everything you solder and assemble to build the finished counter — control-box protoboard, the off-board harnesses (LED rim, 8 piezo wedges, slab DC rail), and the frame-side power. Centered on the control-box protoboard wiring. **Print this**: one printout lives at the bench. Tick the boxes as you go. **Authoritative sources**: `turn_counter_design_doc.md` (full rationale), `shopping_list.md` (sourcing + manufacturer PNs), and the two protoboard SVGs embedded here. Where this guide and those disagree, the SVGs win for wiring.

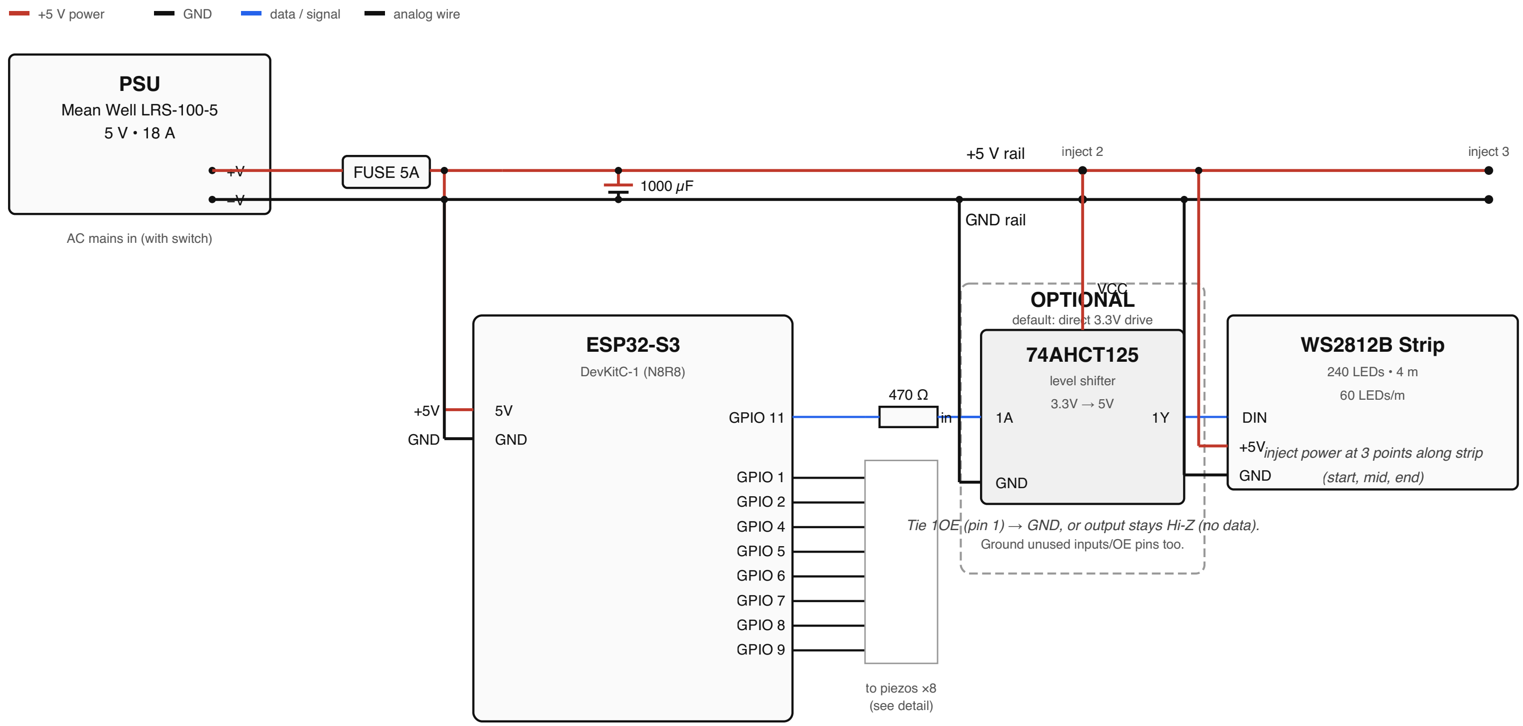
> **How to use this guide**: work top to bottom. Sections 5 → 7 are the soldering; do them in order (control board first — you can bench-test it before committing to the off-board harnesses). Every checklist item is a checkbox. Do not skip Section 8 (pre-power continuity) — it is the difference between "it works" and a dead LED strip.

1. What you're building — signal & power flow

One ESP32-S3 reads 8 piezo discs (one per table side) and drives a single 240-LED WS2812B strip that rings the octagon. The strip is driven directly at 3.3 V through a 470 Ω resistor (a 74AHCT125 level shifter is an optional add-back — §5.2). A frame-mounted 5 V / 18 A PSU feeds the slab through a single Anderson Powerpole DC disconnect so the top lifts off.

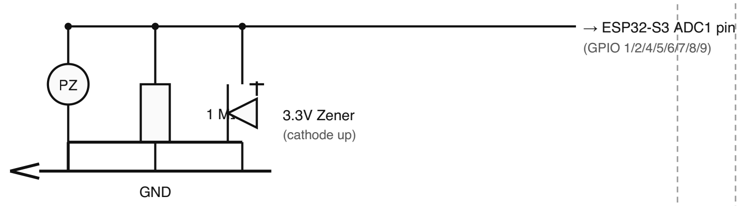
Octagonal Turn Counter — Wiring Diagram

8 piezos • ESP32-S3 • WS2812B (240 LEDs) • 5V/18A PSU



Per-Piezo Input Network (×8)

one of these for each side



Zener clamps spikes from hard taps. 1 MΩ pulls ADC line down so it doesn't float.

Pin Assignments & Notes

ESP32-S3 GPIO	Function
GPIO 11	LED data out → 470 Ω → strip DIN (direct 3.3 V; '125 optional)
GPIO 1	Piezo side 1 (ADC1_CH0)
GPIO 2	Piezo side 2 (ADC1_CH1)
GPIO 4	Piezo side 3 (ADC1_CH3)
GPIO 5	Piezo side 4 (ADC1_CH4)
GPIO 6	Piezo side 5 (ADC1_CH5)
GPIO 7	Piezo side 6 (ADC1_CH6)
GPIO 8	Piezo side 7 (ADC1_CH7)
GPIO 9	Piezo side 8 (ADC1_CH8)

All piezos on ADC1 — works with Wi-Fi/BLE active. Avoid GPIO 0, 3, 19, 20, 45, 46.

Figure 1 — full system wiring. PSU (top-left) feeds the +5V/GND rails; the ESP32-S3 drives 8 piezo ADC inputs and one LED data line through a 470 Ω resistor. The 74AHCT125 level shifter shown in the data path is an OPTIONAL robustness upgrade — the default build drives the strip directly at 3.3 V (§3). Three injection points distribute current along the 240-LED strip.

The five nets that matter at the bench:

Net	Runs	Wire
+5 V	PSU → fuse → Powerpole → slab WAGO node → control board + 3 strip injection points	14 AWG (disconnect), 20 AWG (board pigtail/rails)
GND	common return for everything — PSU, board, strip, all 8 piezos	14 / 20 / 22 AWG
LED data	GPIO 11 → 470 Ω → strip DIN (direct 3.3 V drive; level shifter optional — see §5.2)	22 AWG stranded
Piezo ADC ×8	each piezo → 1 MΩ + Zener **at the disc** → twisted pair → its ADC pin	22 AWG stranded twisted pair
AC mains	wall → IEC inlet → SPST switch (Line only) → PSU	stays on the frame, never crosses the disconnect

Pin map (from `turn_counter.ino` — verify against your DevKitC-1 v1.1 silkscreen):

ESP32-S3 GPIO	Function	Note
11	LED data out → 470 Ω → strip DIN	direct 3.3 V drive
1, 2, 4, 5, 6, 7, 8, 9	Piezo sides 1–8 (ADC1)	all on **ADC1** so tap detection survives Wi-Fi/BLE

Avoid GPIO 0, 3, 19, 20, 45, 46 (strap/USB/PSRAM pins). All 8 piezos are deliberately on ADC1 — ADC2 returns 0 whenever the radio is active.

2. Bill of materials — build view

Full sourcing, quantities, and manufacturer PNs are in `shopping_list.md`. This is the bench-side "is it in front of me" list.

On the control board

✓	Part	Qty	Note
	ESP32-S3-DevKitC-1 (N8R8)	1	socketed on 2× 1×22 female headers — do **not** solder directly
	470 Ω 1/4 W resistor	1	LED data series — spliced **in-line** in the data run, near DIN
	1000 μF electrolytic	1	across 5V/GND at strip entry, **+ to 5V**
	Half-size Perma-Proto (30-col, Adafruit 571)	1	the control board — see `doc-src/protoboard_half_layout.svg`
	JST-XH 3-pin in-line pair	1	strip harness service disconnect (a few inches off-board)
	JST-XH 2-pin in-line pair	8	one per piezo — inline service disconnect, off-board
	74AHCT125 + DIP-14 socket	*(opt)*	*optional level-shifter add-back if pixel 1 glitches (§3) — not built by default*

(Off the board by design: the 8× 1 MΩ + 8× Zener are twisted into the pigtails at the discs (§4.3); the optional 10–47 kΩ ADC series resistors were dropped; direct 3.3 V drive means no level shifter.)

Off-board (slab)

✓	Part	Qty	Note
	WS2812B strip, 60 LED/m, ~4 m	1	cut into 8 side-segments
	Aluminum LED channel + diffuser	4–5 m	22.5° miters at each corner
	27 mm piezo disc	8	one per side, in a rim bore with ~3–5 mm floor (design doc §4.3)
	1 MΩ 1/4 W + 3.3 V Zener (1N4728A)	8 + 8	twisted + soldered into each disc's pigtail, band toward signal
	JST-XH pre-crimped pigtails	9 pr	strip 3-pin + 8× piezo 2-pin in-line disconnects (all off-board)
	WAGO 221 lever connectors	few	branch the slab DC rail
	Project box (Hammond 1591BBK) + M3 standoffs	1	control-box enclosure
	Rubber grommet	1+	slab cable entry (buy first, drill to match)

Frame side

✓	Part	Qty	Note
	Mean Well LRS-100-5 PSU (5 V / 18 A)	1	on a ventilated wood block
	IEC C14 inlet + SPST 20 A rocker switch	1 ea	discrete AC approach
	Inline 5 A blade fuse + holder	1	**DC side**, PSU +V → Powerpole
	Anderson Powerpole 30 A housings + contacts	3 pr	the DC disconnect (solder + heat-shrink)
	14 AWG silicone wire (red + black)	~25 ft	disconnect run, twisted pair

3. Tools & consumables

Essential: temp-controlled iron (Pinecil V2 / Hakko), 63/37 leaded rosin solder, flush cutters, self-adjusting strippers, multimeter (continuity + DC volts). **Strongly recommended**: helping hands / PCB vise, flux pen (Kester 951), heat-shrink assortment, lighter or heat gun, solder wick, silicone bench mat. **Consumables at hand**: 22 AWG stranded hookup wire (6 colors), 20 AWG silicone (red/black), 14 AWG silicone (red/black), heat-shrink, isopropyl for cleanup.

> **Wire gauge rule for this build**: signal = 22 AWG stranded (insulated). Board power runs/pigtail = 20 AWG (14 AWG won't fit protoboard holes; 22 AWG is under the layout's ≥20 AWG spec). DC disconnect run = 14 AWG silicone.

4. Before you solder

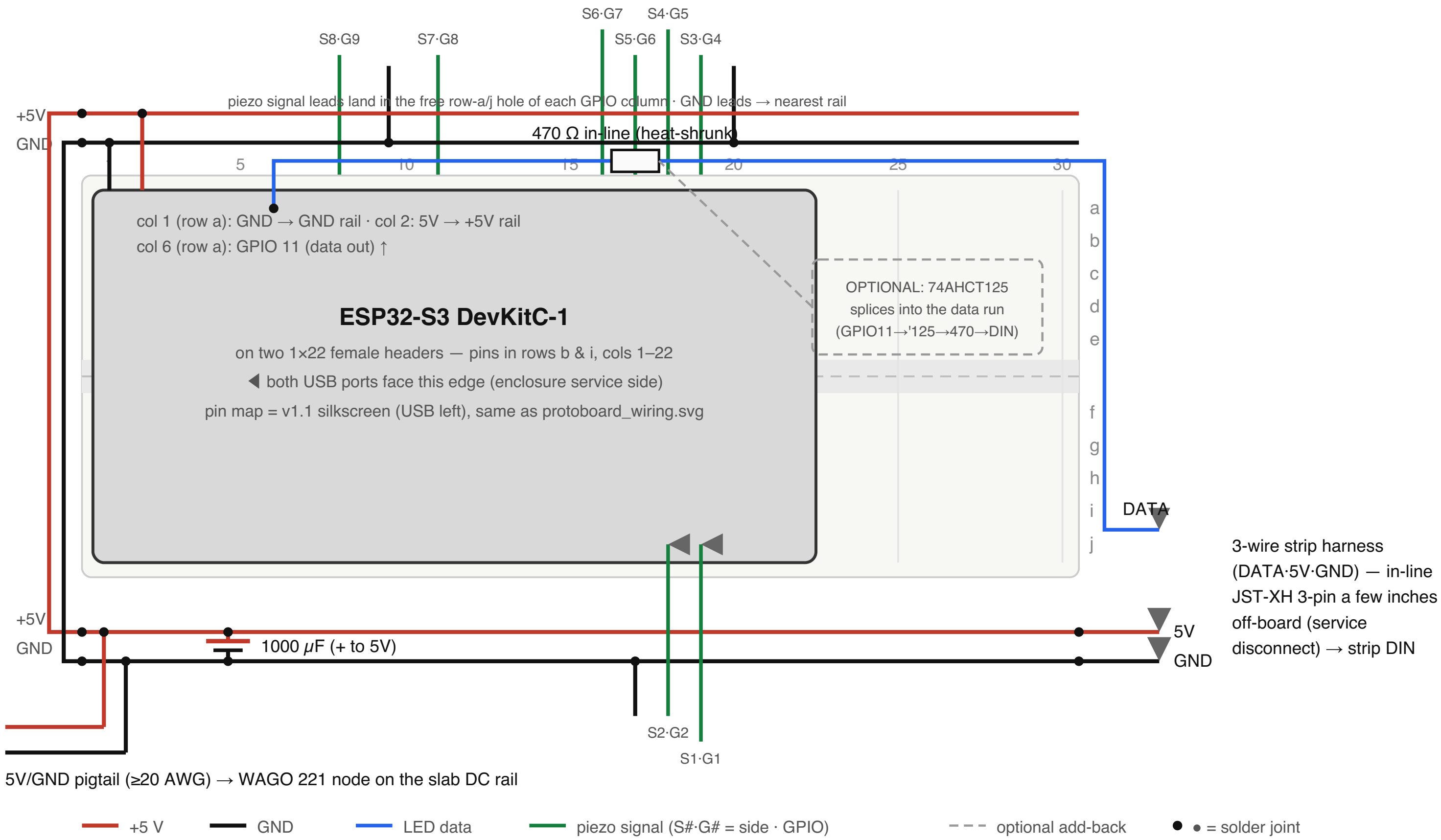
- [ ] Iron tinned and at temp (~330–350 °C for leaded); flux pen within reach
- [ ] Bench mat down; strip is powered **off** and unplugged from any supply
- [ ] Piezos: solder leads in **under 2 seconds** per pad — the discs are heat-sensitive
- [ ] Keep the mains habit from day one: **unplug from the wall before opening the enclosure** (the SPST switch breaks Line only)
- [ ] Have `shopping_list.md` open for the Powerpole solder procedure when you get there

5. Control-box protoboard — the heart of the build

Two diagrams drive this section. The **placement insert** shows where each component sits on the Perma-Proto; the **wiring diagram** is the point-to-point wire list you solder in order. Print both; keep them side by side.

Control-Box Protoboard — Half-Size, Direct-Drive

half-size Perma-Proto (30 cols) · piezo networks at the discs (§4.3) · no on-board resistors, no level shifter, no JSTs · pigtails solder direct



Board: half-size Perma-Proto (Adafruit 571 — 81×46 mm, 30 cols). The ESP32 socket fills cols 1–22; cols 23–30 are spare. Only three things live on the board: the socket, the 1000 µF cap, and the GPIO 11 → 470 Ω → strip data run.

Piezos: each disc carries its own 1 MΩ + Zener (§4.3). The two-wire pigtails solder directly into the row-a/j holes — signal to the GPIO column, GND to the nearest rail. No on-board resistors, no piezo JSTs. Label every pair S1–S8.

LED data: direct 3.3 V drive — GPIO 11 → 470 Ω (in-line, near DIN) → strip. No level shifter. 3.3 V is just under the WS2812B's ~3.5 V high threshold, so the FIRST pixel is the only one at risk; if it ever glitches, add the optional 74AHCT125 into the data run (dashed box) — GPIO 11 → '125 → 470 Ω → DIN, VCC to 5V, 1OE to GND. See design doc §3.

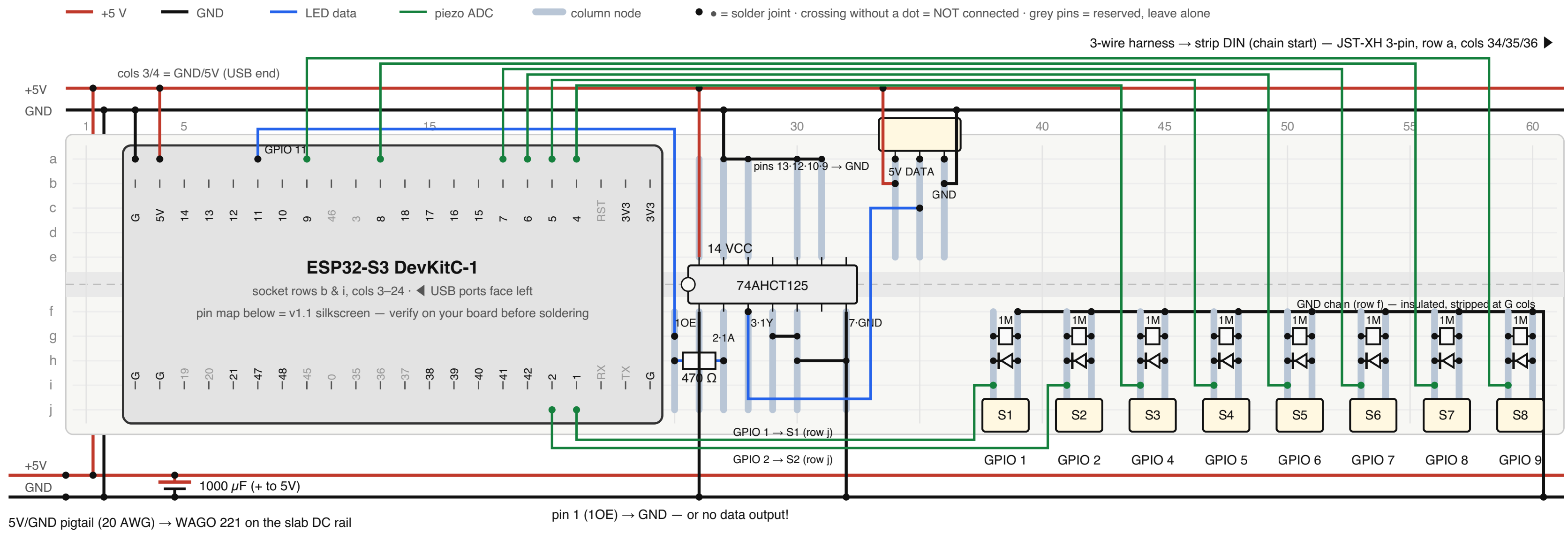
Strip-start share (≤ ~2 A) flows through the pigtail + rails: ≥20 AWG. One common GND for ESP32, strip, and all 8 piezos. Enclosure: the 81 mm board fits a ~112×62×31 mm box (e.g. Hammond 1591B). USB + PSU 5V may coexist, but unplug the strip harness when flashing on USB-only power.

Solder female headers LAST with the DevKit seated as a jig (never solder the module itself).

Figure 2 — control-box protoboard placement (half-size Perma-Proto, 30 cols), direct-drive default. ESP32-S3 socketed at cols 1–22 (USB ports left); cols 23–30 spare. Only the socket, the 1000 µF cap, and the GPIO 11 → 470 Ω → strip data run are on the board — the 1 MΩ + Zener live at each disc and the pigtails solder directly. The dashed box shows where the optional 74AHCT125 would go. Wire runs and solder order are in §5.2.

Control-Box Protoboard — Wiring (Phase 5)

point-to-point companion to the placement insert (protoboard_layout.svg) · pin map = ESP32-S3-DevKitC-1 v1.1, USB ports left · every wire in the list below



WIRE LIST — solder in this order (components are placed per the insert)

- **P1** pigtail red 20 AWG: WAGO +5V → bottom +5V rail (left end)
- **P2** pigtail black 20 AWG: WAGO GND → bottom GND rail (left end)
- **P3** bridge 20 AWG: bottom +5V rail ↔ top +5V rail (before col 1)
- **P4** bridge 20 AWG: bottom GND rail ↔ top GND rail (before col 1)
- **P5** 1000 μF: + → bottom +5V rail, – → bottom GND rail (at entry)
- **P6** col 3 row a (DevKit GND, USB end) → top GND rail
- **P7** col 4 row a (DevKit 5V, USB end) → top +5V rail
- **P8** col 26 row a (IC pin 14, VCC) → top +5V rail
- **P9** col 26 row j (IC pin 1, 1OE) → bottom GND rail
- **P10** col 32 row j (IC pin 7, GND) → bottom GND rail
- **P11** row a: chain cols 27·28·30·31 (IC pins 13/12/10/9), col 27 → top GND rail
- **P12** row g: col 29 ↔ col 30 (pins 4/5) · row h: col 30 ↔ col 32 (to GND)
- **L1** col 34 row b (LED 5V) → top +5V rail, around the JST body
- **L2** col 36 row b (LED GND) → top GND rail, around the JST body
- **D1** col 8 row a (GPIO 11) → col 25 row g · 470 Ω sits col 25→27 row h
- **D2** col 28 row g (IC pin 3, 1Y) → col 35 row c (LED DATA), around the IC
- **A1** S1: col 21 row j (GPIO 1) → col 38 row i
- **A2** S2: col 20 row j (GPIO 2) → col 41 row i
- **A3** S3: col 21 row a (GPIO 4) → col 44 row i
- **A4** S4: col 20 row a (GPIO 5) → col 47 row i
- **A5** S5: col 19 row a (GPIO 6) → col 50 row i
- **A6** S6: col 18 row a (GPIO 7) → col 53 row i
- **A7** S7: col 13 row a (GPIO 8) → col 56 row i
- **A8** S8: col 10 row a (GPIO 9) → col 59 row i
- **G1** insulated GND chain, row f: G cols 39·42·45·48·51·54·57·60 → bottom GND rail (right end)
- **O1** off-board: LED JST → strip DIN (chain start) · S1–S8 JSTs → their wedge piezos (label the cables!)
- **O2** off-board: pigtail lands in the WAGO 221 node on the slab DC rail · leave a USB service loop reachable

All signal wires 22 AWG stranded (insulated); power runs 20 AWG. Row a/j holes are the free holes above/below the socket pins — every other hole in those columns is hidden under the DevKit.
Wire colors follow the legend. Before first power-up: continuity-check +5V↔GND for shorts, then buzz each S column to its GPIO pin and the LED DATA pin to IC pin 3.

Figure 3 — retired full-size wiring diagram, kept for its DevKitC-1 v1.1 pin map and the electrical topology (identical in the half-size build). Column numbers here are the OLD full-size positions — for the current build follow §5.2 and Figure 2; this drawing also shows the piezo networks on-board, which now live at the discs. ₺ = solder joint; a crossing without a dot is NOT connected.

5.1 Place components (per the layout insert)

- ☐ Solder the two 1×22 female header strips (ESP32 socket) at rows b & i, **cols 1–22**, **USB ports facing left**
- ☐ Place the 1000 μF cap (**+ leg to the 5 V rail**)
- ☐ That's the whole board. The piezo networks live at the discs (§6.2), the 470 Ω is an in-line splice in the data wire, and there is no level shifter (direct 3.3 V drive — cols 23–30 stay empty; that's where the optional '125 would go)

5.2 Solder the wire list — in this order

Power & rails first, then LED data, then the piezo pairs. Columns refer to `doc-src/protoboard_half_layout.svg`.

Power in & rails

- ☐ **P1** — pigtail red 20 AWG: WAGO +5V → bottom +5V rail (left end)
- ☐ **P2** — pigtail black 20 AWG: WAGO GND → bottom GND rail (left end)
- ☐ **P3** — bridge 20 AWG: bottom +5V rail ↔ top +5V rail, routed around the board's **left edge** (the socket occupies col 1)
- ☐ **P4** — bridge 20 AWG: bottom GND rail ↔ top GND rail, same route
- ☐ **P5** — 1000 μF: + → bottom +5V rail, **–** → bottom GND rail (at entry)

ESP32 power

- ☐ **P6** — col 1 row a (DevKit GND, USB end) → top GND rail
- ☐ **P7** — col 2 row a (DevKit 5V, USB end) → top +5V rail

LED data path *(direct 3.3 V drive — no level shifter)*

- ☐ **D1** — col 6 row a (GPIO 11) → blue harness lead out the right board edge (strip DATA). The **470 Ω is spliced in-line** in this run, heat-shrunk, as close to the strip DIN as practical
- ☐ *(optional robustness upgrade — skip unless pixel 1 glitches)* insert a 74AHCT125 in this run: GPIO 11 → '125 input → '125 output → 470 Ω → DIN; VCC → +5V, 1OE → GND. See design doc §3

LED strip harness *(direct-soldered — no board JST)*

- ☐ **L1** — red harness lead: bottom +5V rail (right end) → strip harness
- ☐ **L2** — black harness lead: bottom GND rail (right end) → strip harness
- ☐ **L3** — bundle DATA + 5V + GND; fit the **in-line JST-XH 3-pin** a few inches off-board (service disconnect) → strip DIN + start injection

Piezo pairs (22 AWG twisted; signal lands direct on the ADC hole — no series resistor; each pair carries an inline JST-XH 2-pin off-board, labeled S1–S8)

- ☐ **A1** — S1 (GPIO 1): signal → col 19 row j · GND lead → bottom GND rail
- ☐ **A2** — S2 (GPIO 2): signal → col 18 row j · GND lead → bottom GND rail
- ☐ **A3** — S3 (GPIO 4): signal → col 19 row a · GND lead → top GND rail
- ☐ **A4** — S4 (GPIO 5): signal → col 18 row a · GND lead → top GND rail

- ☐ **A5 —** S5 (GPIO 6): signal → col 17 row a · GND lead → top GND rail
- ☐ **A6 —** S6 (GPIO 7): signal → col 16 row a · GND lead → top GND rail
- ☐ **A7 —** S7 (GPIO 8): signal → col 11 row a · GND lead → top GND rail
- ☐ **A8 —** S8 (GPIO 9): signal → col 8 row a · GND lead → top GND rail

Off-board handoff

- ☐ **O1 — label every pair S1–S8 at both ends** before routing — there are no keyed connectors to save you now
- ☐ **O2 —** pigtail lands in the WAGO 221 node on the slab DC rail · leave a USB service loop reachable

> **Row a/j reminder**: those are the free holes above/below the socket pins — every other hole in those columns is hidden under the DevKit. Don't insert the ESP32 until after Section 8 passes.

6. Off-board harnesses (slab)

6.1 LED rim

- ☐ Cut 8 channel pieces, 22.5° miters each end; dry-fit around the slab's outer edge
- ☐ Cut the strip into 8 ~30-LED segments **between pads**; lay into channels, peel adhesive
- ☐ Solder 3-conductor jumpers (5V/GND/Data) across each corner; heat-shrink each joint
- ☐ Continuity-check the data line end-to-end across the assembled strip
- ☐ Solder power-injection pigtails at **3 points**: strip start (DIN), the mid corner (~between sides 4 & 5), and the strip end
- ☐ **Calibrate the per-side LED counts** over serial (tap_light: `0`–`7` select side, `+`/`-` adjust, `p` print) until every color flip lands on a corner — the table persists in NVS and turn_counter reads it automatically; paste the printed array into both sketches' defaults

6.2 Piezo modules (×8)

- ☐ Drill 8 rim bores from the underside, one per side near the outer edge, leaving a **~3–5 mm floor** — never through to the laminate (design doc §4.3)
- ☐ Solder leads to all 8 piezos (red +, black –), under 2 s per pad
- ☐ Twist each disc's 1 MΩ + Zener legs into the pigtail wires ~1/2" off the disc — one soldered bundle per node, **Zener band toward signal** (check before twisting); heat-shrink each node separately. Terminate the disc pigtail in its **inline JST-XH 2-pin**, labeled S1–S8 (signal on pin 1, GND on pin 2 — keep all 8 consistent)
- ☐ Bench-test each: multimeter on AC volts, tap, expect a brief swing
- ☐ Glue each piezo into its bore, brass face to the wood floor (CA or thin epoxy); vacuum the dust first; press 30 s
- ☐ Route the twisted pairs through the underside kerfs to the control box; **label each pair to its side (S1–S8)**

6.3 Slab DC rail & disconnect

- ☐ Drill the grommet hole (buy grommet first, drill to match); run the slab-side DC cable in
- ☐ Solder Powerpole contacts to the slab-side cable ends; insert into housings (red +V, black GND)
- ☐ Branch the slab DC cable into a WAGO 221 node → control-box 5V/GND **and** the 3 strip injection points
- ☐ **Verify polarity at every junction with a multimeter** — reversed 5 V instantly kills the strip
- ☐ P-clip / cable mount within 4" of the grommet — strain relief hits the clip, not the connector
- ☐ Land the 8 labeled piezo pairs on the board (per §5.2 A1–A8), then click the 8 piezo JSTs and the strip harness's in-line JST together

7. Frame side (permanent) — AC mains & PSU

> **Mains safety**: do this section with everything unplugged from the wall. AC never crosses the disconnect — only +5 V DC does. Verify all AC continuity with a meter *before* the plug ever goes in the wall.

- [ ] Mount the PSU to a ~4×6×1" wood block; screw the block to a **ventilated** spot under the frame
- [ ] Wire AC: wall plug → IEC C14 inlet → SPST switch (Line only) → PSU **L**. Neutral runs inlet → PSU **N** direct (no switch). Earth runs inlet Earth → PSU chassis lug (no switch). 5 faston connections total (3 Line + 1 N + 1 Earth)
- [ ] Wire PSU **+V** → **inline 5 A fuse** → **Powerpole +V**; PSU **–V** → **Powerpole –V** direct
- [ ] Solder Powerpole contacts (procedure in `shopping_list.md`), insert red=+V / black=GND, slide housings together
- [ ] Leave ~18" of slack so you can lift the slab before disconnecting
- [ ] Continuity check: switch ON → Line continuous inlet→PSU; switch OFF → Line open; N & Earth always continuous
- [ ] **Frame-only power test** (no slab): switch on, meter the Powerpole → expect +5 V red-to-black. Switch off.

8. Pre-power-up continuity checklist — DO NOT SKIP

Do all of these with the board **unpowered** and the ESP32 still **out of its socket**, before the first power-up.

- ☐ **+5 V ↔ GND**: probe for shorts. Expect open (the cap may give a brief charge beep, then settle open). A hard short here = find it before you apply power
- ☐ **Buzz each S column → its GPIO pin** (S1→GPIO1 … S8→GPIO9), with the piezo JSTs mated — confirms the ADC jumpers A1–A8 through the connectors
- ☐ **LED DATA → IC pin 3 (1Y)** continuous — confirms D2
- ☐ **1OE (IC pin 1) → GND** continuous — confirms P9 (or the strip stays Hi-Z / dark)
- ☐ **IC pin 7 (GND) → GND** and **IC pin 14 (VCC) → +5 V** — confirms P8/P10
- ☐ **1000 µF polarity**: + leg on the 5 V rail
- ☐ Only now: seat the ESP32-S3 into its socket, orientation correct (plus the 74AHCT125 if you fitted the optional shifter)

9. Firmware flash

- [ ] Arduino IDE: install the **ESP32 board package**; select **ESP32-S3-DevKitC-1**
- [ ] **Partition Scheme → Minimal SPIFFS (min_spiffs)** — `turn_counter.ino` is ~9 KB too big for the default 1.25 MB app partition and won't link otherwise
- [ ] **USB CDC On Boot → Enabled**; upload speed 115200 if it's flaky
- [ ] Port: pick the **`usbserial-*` (CP2102)** device — the `usbmodem` ports are not the board
- [ ] Confirm `PIEZO_PINS[] = {1,2,4,5,6,7,8,9}` and `NUM_LEDS` / `LEDS_PER_SIDE` match the installed strip
- [ ] Flash, open Serial Monitor at 115200; watch the per-side baseline seed lines at boot
- [ ] If it won't enter download mode: hold **BOOT**, tap **RESET**, release **BOOT**
- [ ] Piezo map: run `make map-piezos`, tap each side as it lights white, then power-cycle and confirm taps land on the right seats. Do this instead of chasing a crossed harness back through the wiring
- [ ] OTA (optional): copy `secrets.example.h` to `firmware/turn_counter/secrets.h` and fill it in, `make flash-turn` once over USB, confirm `ping turn-counter.local` resolves, then confirm `make ota` completes
- [ ] Phone UI: open `http://turn-counter.local/` (iPhone) and `http://<ip>/` (Android), confirm the mode buttons, brightness slider and on/off all work, and that the status follows taps made at the table
- [ ] Guest access: DHCP-reserve the board's IP (its MAC is printed at boot), then `make qr URL=<ip>` and stick the printed sticker under the table

10. First power-up & smoke test

- ☐ **Slab bench test** (before mating to the frame): connect the bench-test Powerpole pigtail + a 5 V supply → confirm the strip lights and taps advance the lit zone
- ☐ Disconnect the bench supply; **mate slab Powerpole → frame Powerpole**
- ☐ First full power-up: switch on, **watch for smoke** (no smoke = good)
- ☐ Tap through every player position — the lit zone should sweep cleanly across all 8 sides; look for dead LEDs, wrong colors, corner gaps
- ☐ Test the disconnect: off → unmate → lift the slab → set back → reconnect → on → confirm state restored

11. Bench troubleshooting

Symptom	Likely cause	Fix
Whole strip dark	switch off, blown fuse, reversed polarity, or **1OE not grounded**	check switch → fuse → PSU output → polarity → P9
First LED wrong color / flicker	weak data into pixel 1 (3.3 V direct is marginal)	confirm 470 Ω + common ground (D1); if it persists, add the optional 74AHCT125 (§5.2)
Far-end LEDs tint pink/orange	voltage droop	check mid/end power injection
Constant/chaotic taps regardless of piezo, rail-to-rail ADC with a stable baseline	**floating ADC pin — lost GND return** (broken ground), not a threshold problem	trace the GND chain (G1) and the piezo/board common ground; it's a broken ground, chase continuity
Tap doesn't register	`TAP_DELTA` too high, bad piezo joint, glue not contacting	lower `TAP_DELTA`, reflow, re-glue
Tap on one side lights another	piezo cable mapping wrong	check `PIEZO_PINS[]` order vs physical wiring
Won't flash via USB	wrong port / CDC / speed	use `usbserial-*` port, USB CDC On Boot = Enabled, drop to 115200, BOOT+RESET
Solder joint dull/blobby	cold joint	reflow with flux, hold ~2 s while cooling

12. Quick reference

Pin map: LED data = **GPIO 11** → 470 Ω → strip DIN (direct 3.3 V; optional 74AHCT125 in between). Piezos = **GPIO 1, 2, 4, 5, 6, 7, 8, 9** (ADC1), sides 1–8 in order. **Firmware knobs**: `NUM_LEDS` (match installed count), `LEDS_PER_SIDE = NUM_LEDS/8` rounded down, `TAP_DELTA` ≈ 30% of a deliberate tap's jump above baseline, `DEBOUNCE_MS = 250`. **Golden rules**: 1OE → GND or no light · + to 5 V on the cap · polarity-check every DC junction · unplug the wall before opening the box · label every off-board cable.

Companion to `turn_counter_design_doc.md`. Wiring per `doc-src/protoboard_wiring.svg`.